

How the service works today

owis_find_deal_engine — search service after the local tests of 6–7 October 2026: sources per shop, settings, costs and limits.

In one sentence: an app sends a product title and a country; the server checks the login, answers from its cache when it can (free), otherwise asks each shop of that country in parallel — Amazon and eBay through SerpApi, AliExpress, Temu and SHEIN through Apify scrapers — and returns one list with prices and direct product links.

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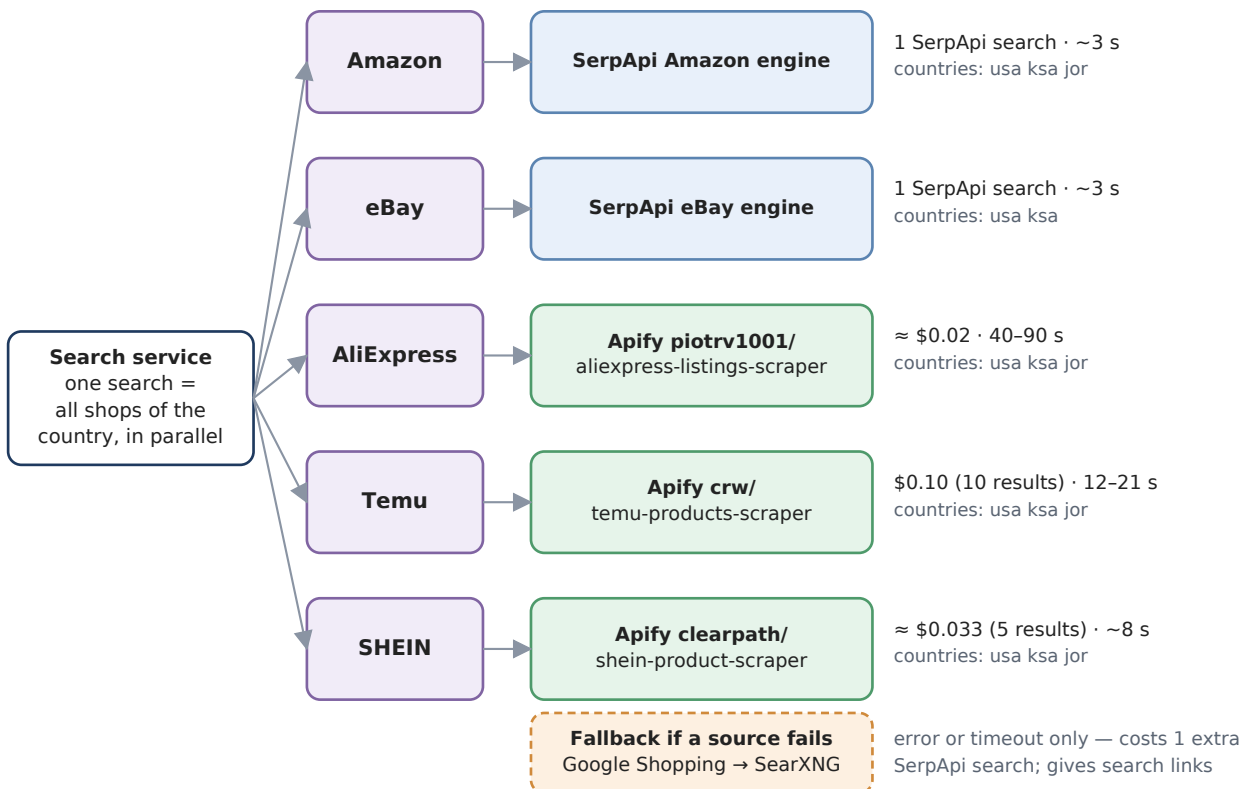
1. Countries and shops

Only the shops that deliver to the country are searched (file `internal/markets/markets.json`).

Country	Code	Amazon	eBay	AliExpress	Temu	SHEIN
United States	usa	amazon.com	ebay.com	✓	✓	us.shein.com
Saudi Arabia	ksa	amazon.sa	ebay.com	✓	✓	ar.shein.com *
Jordan	jor	amazon.com	—	✓	✓	ar.shein.com *

* The SHEIN scraper reads the US site only, so links and prices are from us.shein.com in every country (accepted).

2. Where each shop's results come from



Each shop has its own source; all shops of a country are searched at the same time.

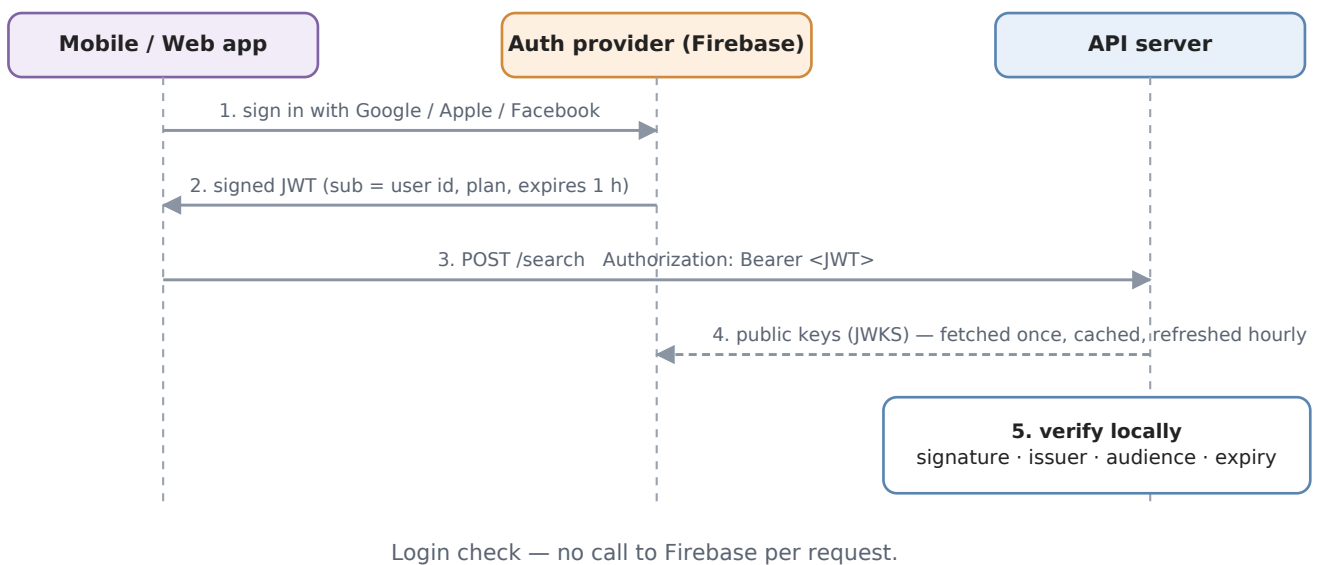
Shop	Source	What we get	Link format
Amazon	SerpApi Amazon engine	price, rating, delivery, sponsored mark	amazon.<tld>/dp/<ASIN>
eBay	SerpApi eBay engine (new 7 Oct)	price (lowest of a range), condition, shipping, rating	ebay.com/itm/<id>
AliExpress	Apify piotrv1001/aliexpress-listings-scraper	price (US, no currency → set to USD), image	aliexpress.com/item/<id>.html
Temu	Apify crw/temu-products-scraper	price, image	temu.com/goods.html?goods_id=<id>
SHEIN	Apify clearpath/shein-product-scraper	sale price (not the old price), image	us.shein.com/...-p-<id>.html

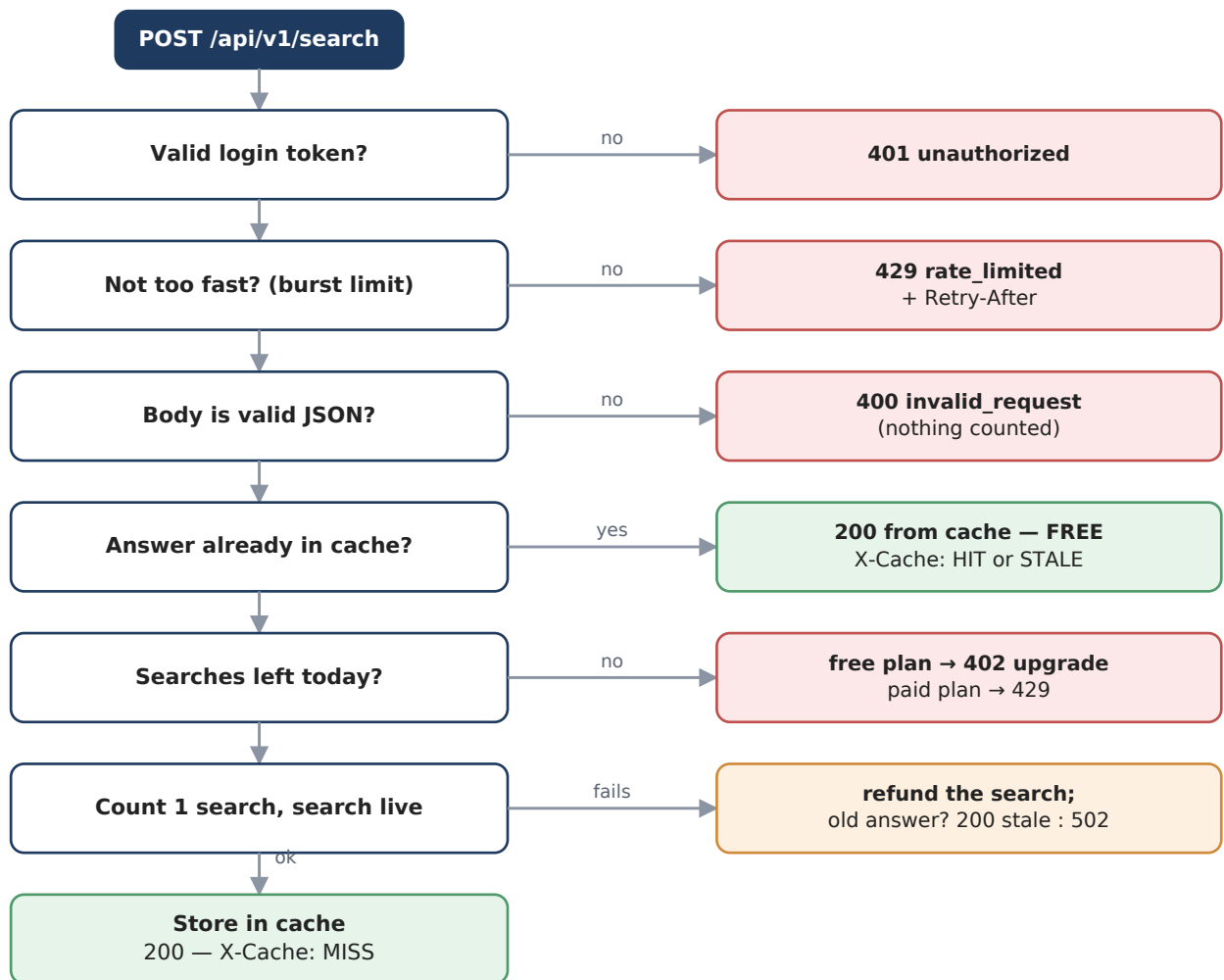
Rules applied to every result

- **Relevant only:** at least two thirds of the search words must appear in the title (shops also return merely related items).
- **Direct product pages only;** links are cleaned (tracking parameters and country prefixes removed).
- **No duplicates** within a shop (the cheapest copy is kept).
- Results whose title is only the shop's name (e.g. "Temu") are dropped.
- **Fallback** only on error or timeout: Google Shopping (SerpApi) → SearXNG (free). A scraper that finds nothing relevant means "no results" for that shop (APIFY_EMPTY_FALLBACK=false) — no extra time or cost.
- A source that keeps failing is skipped for a minute (circuit breaker).

3. The request flow

Login: the app signs the user in with Firebase and sends the token with every search; the server checks it locally with Firebase's public keys.



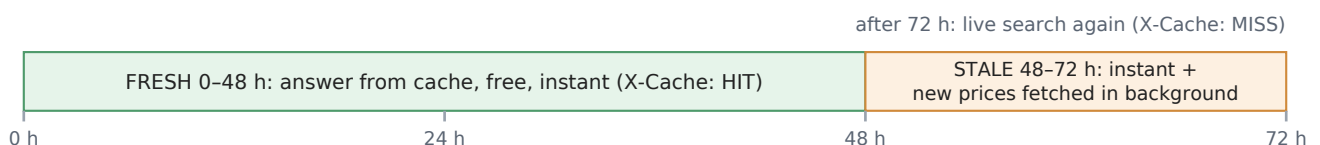


POST /api/v1/search — every check happens before any paid search is made.

- **Burst limit:** 1 request per second per user, bursts of 5 (RATE_LIMIT_RPS, RATE_LIMIT_BURST).
- **Daily allowance:** free plan 20 live searches, pro 500 (PLANS=free:20, pro:500). Cached answers are not counted. Failed searches are refunded.
- **Time:** a live search waits for the slowest shop: usually 25–50 s (AliExpress is the slowest). Each scraper gets up to 90 s; the whole search up to 100 s.

4. Cache: 48 hours fresh

Each answer is stored per country + title (case and spaces ignored), in Redis.



A shop failed → that answer is fresh only 10 min · no results → 30 min · pull-to-refresh at most once per 10 min per search.

New defaults since 7 Oct: CACHE_FRESH_TTL=48h, CACHE_STALE_TTL=72h.

Effect: a search is paid for at most once every 48 hours, whoever searches it. Prices can be up to 2 days old — the app shows them as "from \$X · approx." and the link opens the real price.

5. Costs and limits

Item	Cost per live search	Notes
Temu (Apify)	\$0.100	charges for 10 results even if fewer are kept → keep 10
SHEIN (Apify)	≈ \$0.033	5 results × ≈ \$0.0066
AliExpress (Apify)	≈ \$0.02	estimate from the monthly total
Apify total	≈ \$0.155	free plan: \$5 credit / month
SerpApi	jor: 1 search · usa / ksa: 2	Amazon (+ eBay); free plan 250 searches / month
Fallback (rare)	+1 SerpApi search	only when a source fails or times out
Cached answer	\$0	for 48 h

Safety limits on spending

- APIFY_MAX_CHARGE_USD=0.20 — the most one scraper run may cost (needed by the SHEIN scraper, which is billed per event).
- APIFY_<SHOP>_MAX_ITEMS — results per shop (SHEIN 5; Temu 10).
- Daily allowance per user (section 3) and the cache (section 4).
- Console test runs in Apify have **no** limit unless set — keep the result count small there.

Apify platform prices (for comparing scrapers later)

Scrapers with their own price per result (Temu, SHEIN) include everything. Only scrapers billed "per usage" add platform costs: \$0.20 per compute unit (1 GB × 1 hour; a 60 s run with 1 GB ≈ \$0.003), residential proxy \$8 / GB (avoid), datacenter proxy included. Storage and data transfer are negligible for us.

6. The answer format

```
{
  "query": "usb c hub", "country": "ksa",
  "results": [
    {
      "market": "ebay", "title": "USB C Hub 4K HDMI Adapter ...",
      "link": "https://www.ebay.com/itm/266881403243",
      "thumbnail": "https://i.ebayimg.com/...", "price": 13.99, "currency": "$",
      "position": 1, "provider": "serpapi-ebay",
      "extensions": ["Brand New", "Free delivery in 2-4 days"]},
    ...
  ],
  "markets": {"amazon": "ok", "ebay": "ok", "aliexpress": "ok", "temu": "ok", "shein": "ok"},
  "took_ms": 48742, "fetched_at": "2026-10-07T11:29:13Z", "cached": false, "stale": false
}
```

- Results are interleaved by position: the first result of every shop, then the second, ...
- provider: serpapi-amazon, serpapi-ebay, apify, serpapi (Google Shopping fallback), searxng (free fallback). Fallback results may have "link_type": "search".
- markets: per shop ok / no_results / failed.
- Headers: X-Cache (HIT / STALE / MISS), X-Plan, X-RateLimit-Limit / -Remaining / -Reset.

7. Current configuration (.env)

Values used in the local tests, besides login, secrets and keys (never in the repository).

```
SEARCH_PROVIDERS=serpapi,searxng          # fallback chain
SERPAPI_ENGINE=google_shopping            # fallback engine
# SERPAPI_MARKETS defaults to amazon,ebay when SERPAPI_KEY is set

APIFY_MARKETS=aliexpress,temu,shein
APIFY_ALIEXPRESS_ACTOR=piotrv1001/aliexpress-listings-scraper
APIFY_ALIEXPRESS_INPUT={"maxResults":{{max}}, "searchQueries":["{{query}}"],
                        "proxyConfiguration":{"useApifyProxy":true}}
APIFY_ALIEXPRESS_CURRENCY=USD              # scraper returns "currency": null
APIFY_TEMU_ACTOR=crw/temu-products-scraper
APIFY_TEMU_INPUT={"keyword":"{{query}}", "max_items":10, "region":"US",
                  "sort":"relevance"}      # scraper needs >= 10
APIFY_TEMU_MAX_ITEMS=10
APIFY_SHEIN_ACTOR=clearpath/shein-product-scraper
APIFY_SHEIN_INPUT={"enrichDetails":false, "maxItemsPerSearch":{{max}},
                  "quickShip":false, "searchTerms":["{{query}}"], "site":"us",
                  "sortBy":"recommend", "category":""}
APIFY_SHEIN_MAX_ITEMS=5
APIFY_TIMEOUT=90s
APIFY_MAX_CHARGE_USD=0.20
APIFY_EMPTY_FALLBACK=false

CACHE_FRESH_TTL=48h
CACHE_STALE_TTL=72h
PLANS=free:20,pro:500
```

In phase 2 these settings move into the admin panel (applied without restart); .env keeps only start-up values and secrets.

8. Known limits (accepted)

Limit	Why	In the app
Prices are US prices	the scrapers read the US sites; the page in Jordan / Saudi may show another price, currency or discount	show "from \$X · approx."; the link shows the real price
AliExpress welcome deals	many prices (\$0.33, \$1.09) are new-shopper offers	note: "AliExpress prices may be new-shopper offers"
SHEIN on the US site	the scraper supports only us	check another SHEIN scraper later
First live search is slow	25–50 s (scrapers start a browser)	loading bar; repeat searches are instant
Up to 48 h old prices	cache keeps costs low	pull-to-refresh for a new price

9. Test results (6–7 Oct 2026)

Search	Country	Result	Time	Cost
usb c ssd enclosure	usa	5 shops ok; SHEIN 8 via Apify; AliExpress fell back (60 s timeout)	67 s	3 SerpApi + Apify
iphone 18 max case	jor	4 shops ok, no fallback (timeout 90 s)	36 s	1 SerpApi + Apify
usb c hub	ksa	eBay 10 direct links ✓; Temu failed (max_items 5) → fixed	49 s	3 SerpApi + Apify

Search	Country	Result	Time	Cost
fast charger type c	jor	4 shops ok, no fallback	26 s	1 SerpApi + $\approx$ \$0.16

Links opened by the owner: Amazon, eBay, SHEIN and Temu show the same price as returned (Temu / AliExpress pages in Jordan may show local offers).